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Studierichting: Informatica

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$$\begin{aligned} & \lim_{x \rightarrow \infty} (\sqrt{2x^2 + 3} - x) \\ & \lim_{x \rightarrow -\infty} (\sqrt{2x^2 + 3} - x) = +\infty - (-\infty) = +\infty \\ & \lim_{x \rightarrow +\infty} \frac{(\sqrt{2x^2 + 3} - x)(\sqrt{2x^2 + 3} + x)}{(\sqrt{2x^2 + 3} + x)} \\ & = \lim_{x \rightarrow +\infty} \frac{2x^2 + 3 - x^2}{(\sqrt{2x^2 + 3} + x)} \\ & = \lim_{x \rightarrow +\infty} \frac{x^2 + 3}{(\sqrt{2x^2 + 3} + x)} = \lim_{x \rightarrow +\infty} \frac{x^2}{(\sqrt{2} + 1)x} = +\infty \end{aligned}$$

References